

Aditya Sugandhi

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SUMMARY

Senior Full-Stack Software Engineer with 5+ years building **scalable support platforms** and **developer tools**. Expert in **TypeScript (React, Next.js)**, **Python**, and **distributed systems** with production experience in **observability**, **debugging infrastructure**, and **API development**. Delivered **37 RESTful endpoints** serving **35K+ devices** with **99.5% availability** and built **monitoring dashboards** reducing **MTTR by 60%**.

TECHNICAL SKILLS

- **Frontend:** React, TypeScript, Next.js, JavaScript, HTML/CSS, Cloudflare Workers, Component Libraries
- **Backend:** Python, FastAPI, Node.js, Java/Spring Boot, gRPC, RESTful APIs, GraphQL, Microservices Architecture
- **Observability & Debugging:** Grafana, Prometheus, CloudWatch, Distributed Tracing, Log Aggregation, Alerting Systems, Performance Monitoring
- **Data & Infrastructure:** PostgreSQL, Redis, MongoDB, Apache Kafka, AWS (EKS, Lambda, S3, CloudWatch), Kubernetes, Docker
- **Developer Tools & SDKs:** CLI Development, SDK Design, API Documentation, Developer Experience, Internal Tooling
- **Practices:** Incident Response, Root Cause Analysis, Agile/Scrum, Cross-functional Collaboration, On-call Support, SLA/SLO Management

WORK EXPERIENCE

Software Engineer II

Florida State University

Nov 2024 – Present

Tallahassee, FL

- Built **support platform infrastructure** with **37 RESTful API endpoints** and **real-time WebSocket streaming** serving **35K+ IoT devices**; implemented **debugging tools** and **diagnostic APIs** enabling rapid troubleshooting across distributed systems.
- Established **Grafana monitoring dashboards** and **alerting systems** for operational excellence; built **observability pipelines** tracking **500K+ events/day** with automated incident detection reducing investigation time by **40%**.
- Developed **internal SDKs and tool connectors** enabling cross-system integrations; standardized **API contracts** and **developer documentation** improving engineering team productivity and reducing onboarding time.
- Implemented **fault-tolerant workflows** with **Redis caching**, **idempotent retry logic**, and **state management**; achieved **99.2% system reliability** across **172+ networks** with automated health monitoring.
- Collaborated with cross-functional teams to design **debugging interfaces** and **support tools**; provided **technical guidance** on API design patterns and best practices for distributed systems.

Software Engineer I

Aspire Systems

Oct 2020 – Jul 2022

Chennai, India

- Led **full-stack platform development** using **Java/Spring Boot** backend and **React/TypeScript** frontend; architected **microservices** processing **500K+ events/day** with **Kafka** event streaming and **99.5% availability**.
- Built **comprehensive observability infrastructure** with **Prometheus/Grafana** and **CloudWatch**; implemented **distributed tracing** and **log aggregation** reducing **MTTR by 60%** through automated alerting and real-time debugging dashboards.
- Designed **internal developer tools** and **support APIs** enabling cross-system integrations; created **debugging utilities** and **diagnostic endpoints** for rapid incident response and troubleshooting.
- Provided **on-call support** and **incident response** for production systems; conducted **root cause analysis** and implemented preventive measures improving system stability and customer satisfaction.
- Mentored **5 engineers** on **microservices architecture**, **API design**, and **debugging best practices**; established coding standards and review processes for engineering team.

Machine Learning Researcher

Prof. Olmo Zavala Romero — FSU

Jan 2023 – Aug 2024

Tallahassee, FL

- Built **data processing pipelines** handling **1TB+ datasets** using **Python** and **Spark**; implemented **observability dashboards** (Grafana, MLflow) for quality monitoring and performance tracking.
- Developed **workflow orchestration systems** on **Kubernetes** with optimized **retry patterns** and **state management**; achieved **50% pipeline runtime reduction** with **99% job success rate**.
- Created **Python SDKs** and **developer tools** for data processing; collaborated with cross-functional teams to iterate from experimentation to production deployment.

Associate Software Engineer

IMPETUS Technologies

Jun 2019 – Sep 2020

Noida, India

- Developed **Python and TypeScript** backend services for real-time data processing; built **RESTful APIs** handling **100K+ daily requests** with **Redis caching** for low-latency responses.
- Implemented **workflow automation tools** and **internal SDKs** for data pipeline orchestration; created **debugging utilities** supporting engineering team productivity.
- Worked with **PostgreSQL** and **MongoDB** for data storage with focus on **idempotent operations**, **retry logic**, and **data quality monitoring**.

PROJECTS

- **Developer CLI & SDK Platform** — Developed **TypeScript-based SDK** and **CLI tools** for API integration enabling autonomous operations. Built **function-calling interfaces** with **retry logic**, **idempotency**, and **error handling** for enterprise tooling. Tech: **TypeScript, Node.js, REST APIs, Developer Tools**.
- **Full-Stack Support Platform** — Built **AI-powered application** with **TypeScript/Next.js** frontend and **Python/FastAPI** backend. Implemented **real-time workflow orchestration**, **API integrations**, and **monitoring dashboards**. Deployed on **AWS Lambda** with **DynamoDB** and **S3**. Tech: **TypeScript, React, Next.js, Python, AWS, Observability**.
- **Multi-Service Orchestration System** — Architected **21-node workflow system** for automated pipelines. Built **tool connectors** for 5+ external APIs, implemented **conditional routing**, **state management**, and **observability monitoring**. Tech: **Python, FastAPI, Distributed Systems, Monitoring**.

EDUCATION

Florida State University M.S. in Computer Science	Tallahassee, FL
• Relevant Coursework: Distributed Systems, Advanced Database Systems, Machine Learning, Data Communications	
SRM Institute of Science & Technology B.Tech. in Computer Science	Chennai, India

PUBLICATIONS & CERTIFICATIONS

- Review Classification & False Feedback Detection (IJAST) | AWS Solutions Architect (Udemy, 2024)